

Assignment 10

Due Sunday November 23rd, 2025, Self Grading Due Sunday December 7th, 2025

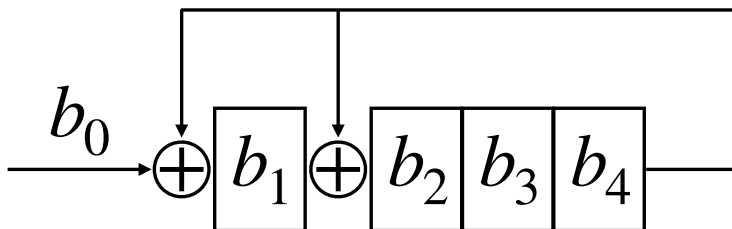
1. Review Andrews Coding Notes
2. Consider a repetition code rate of $r = 1/4$ with PAM2, and $\text{SNR} = 6\text{dB}$.
 - (a) What's the bit error probability without coding?
 - (b) Draw the output of the Coder assuming an input bit sequence of 010.
 - (c) The receiver uses the following rules for determining the bit.

$$\sum_{i=0}^3 b_i \geq 2 \quad \rightarrow \quad \text{bit is a 1}$$

$$\sum_{i=0}^3 b_i < 2 \quad \rightarrow \quad \text{bit is a 0}$$

What's the bit error probability assuming that 1111 and 0000 are transmitted with equal probability? How does it compare without coding?

- (d) What's the coding gain?
3. A CRC-4 generator is implemented using the following shift registers and XOR operations.



- (a) What is the polynomial that represents the generator?
 - (b) What's the probability of a miss detection and false alarm? Assume the bits are iid Bernoulli and $P_e = 10^{-4}$.
4. **Self-grade Homework 9.** Fill out the self-grade rubric for Homework 9. Attach to Homework 10 submission.